

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Fifth Semester B.E. (E&C) (Credit System) Degree Examinations**

January 2023

18ECE61 – ADVANCED SIGNAL PROCESSING

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

Unit – I**Marks BT* CO* PO***

1. a)
- $x[n] = \{-2, 1, -1, 3, 0, 2, 3, -5\}$
- . Evaluate

i. $X(0)$

ii. $X(4)$

iii. $\sum_{k=0}^7 X(K)$

iv. $\sum_{k=0}^7 |X(K)|^2$

10 L*3 1 1

- b) List and explain the properties of complex cepstrum.

10 L2 2 1

2. a) The complex cepstrum,
- $\hat{x}[n]$
- , of a sequence
- $x[n]$
- is the inverse DTFT of the complex logarithm of the DTFT; i.e.,

$$\hat{X}(e^{j\omega}) = \log |X(e^{j\omega})| + j \arg [X(e^{j\omega})].$$

Show that the (real) cepstrum, $c[n]$, defined as the inverse Fourier transform of the log magnitude, is the even part of $\hat{x}[n]$.

10 L4 1 2

- b) Starting with the general form of a Z domain representation of a signal
- $x[n]$
- , derive the expressions for the complex cepstrum
- $\hat{x}[n]$
- .

10 L2 2 1

3. a) Determine the complex cepstrum of the sequence

$$x[n] = x_1[n] * x_2[n]$$

where

$$x_1[n] = a^n u[n] \text{ and } x_2[n] = \delta[n] + b\delta[n+1]$$

10 L3 1 1

- b) Consider a finite-length minimum-phase sequence
- $x[n]$
- with complex cepstrum
- $\hat{x}[n]$
- , and a sequence
- $y[n] = \alpha^n x[n]$
- with complex cepstrum
- $\hat{y}[n]$
- .

i. If $0 < \alpha < 1$, how will $\hat{y}[n]$ be related to $\hat{x}[n]$?ii. How should α be chosen so that $y[n]$ would no longer be minimum-phase?iii. How should α be chosen so that $y[n]$ is maximum-phase?

10 L3 2 1

Unit – II

4. a) Derive Wiener Hopf equations for FIR Wiener filter.

10 L2 3 1

- b) Consider the autocorrelation function of an AR(1) process
- $d(n)$

given by $r_d(k) = \alpha^{|k|}$

Suppose that $d(n)$ is observed in the presence of uncorrelated white noise, $v(n)$ that has a variance of σ_v^2 ,

$$x(n) = d(n) + v(n)$$

Design a first order Wiener filter to estimate the signal $d(n)$.

10 L3 4 1

5.	a)	Explain how adaptive filters are used for noise cancellation.	10	L2	3	1
	b)	Consider a four step predictor using a first order filter $W(z) = w(0) + w(1)z^{-1}$				
	i.	What are the Weiner Hopf equations for the Weiner four step predictor to estimate $x(n+4)$?				
	ii.	If the values of $r_x(k)$ for lags $k = 0$ to 5 are $r_x = [2, 0, -0.3, 0.1, 0.8, 0.6]^T$ Obtain the filter coefficients of the first order Weiner filter.	10	L3	4	1
6.	a)	Derive the expression for minimum Least Square Error in the RLS Algorithm.	10	L2	3	1
	b)	Explain the concept of interpolator. Establish the relationship between the output and input sequences in Z domain for an L-fold expander.	10	L2	4	1
Unit – III						
7.	a)	Explain the concept of DFT polyphase analysis filter banks.	10	L2	5	1
	b)	Explain the principle of Interpolated FIR (IFIR) filters in detail.	10	L2	5	1
8.	a)	With help of examples, explain the basic principles of polyphase decimators and interpolators.	10	L2	5	1
	b)	What are Polyphase Quadrature Mirror Filter (QMF) banks? Explain.	10	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
